

Cheng Yuan (Ross) King

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CAREER ASPIRATION

I am a software engineer and MSc Artificial Intelligence candidate specialising in the testing and reliability of AI systems. I design and release open-source software that automatically tests AI and machine-learning models for failures before they reach users, and data platforms that keep large, inconsistent datasets accurate. I am skilled in Python and SQL, automated testing and continuous integration, and building and deploying services with FastAPI, Docker and cloud data tools.

EDUCATION

University of Sheffield

MSc Artificial Intelligence

Sep 2025 – Sep 2026 | Sheffield, UK

University of Sheffield International College

Pre-Master's Computing and Computer Science

Jan 2025 – Aug 2025 | Sheffield, UK

Queen's University Belfast

BSc Computer Science

Sep 2021 – Jun 2024 | Belfast, UK

SKILLS

Languages : Python, SQL, Bash, Git

AI & Evaluation : LLM evaluation, AISI Inspect, red-teaming, prompt-injection testing, RAG evaluation, agent tooling (MCP)

Testing & Release : pytest, CI/CD, GitHub Actions, Docker, mypy (strict), PyPI packaging

Services & APIs : FastAPI, REST/JSON APIs

Data : dbt, PySpark, Airflow, BigQuery

Spoken : English (Fluent), Mandarin Chinese (Native), Japanese (JLPT N1)

PROJECTS

Agent Release Gates | *Python, Inspect, FastAPI, Docker, CI*

- Built a release gate for LLM agents that replays **8** recorded incidents against any changed agent, prompt, model or tool policy and returns `ship` / `warn` / `block` — so an engineering team catches a behavioural regression in CI instead of after users hit it.
- Found and published the failure that mattered: the retriever scores **99.31%** on the project's own benchmark but **79.92%** on **640** external public cases (TechQA, WixQA; 510 documents). The in-house benchmark is circular by construction, so the external number is the one reported.
- Shipped end to end — CLI on PyPI, Inspect (UK AISI) evaluation suite, FastAPI service, Docker, CI and **309** tests — with a reviewer dashboard as the single evidence surface.

Red-Team Foundry | *Python, Inspect, PyPI*

- Tested whether the public jailbreak benchmarks the field still cites can actually tell models apart: ran **1,883** prompts from **4** pinned adversarial corpora across **12** evaluation cells (2 target models × 2 attack families × up to 4 defence stacks), with bootstrap confidence intervals and per-run API cost.
- Attack success came out at **0–4%** — a negative result — so proved the harness was not simply under-testing: a known-vulnerable model through the identical pipeline scored **80%** attack success, with two independent LLM judges agreeing at $\kappa = +0.935$.
- Reported where the agreement metric collapses instead of banking it — in 11 of 12 cells both judges labelled every case identically, which makes κ undefined rather than perfect — and wrote a CI check that fails the build if that caveat is dropped. **340** tests; published to PyPI; exports to Inspect logs.

TfL Cycle-Hire Data Platform | *PySpark, dbt, DuckDB, Airflow, MCP*

- Unified **41.4M** London cycle-hire journeys from **148** files across **5** incompatible schema eras into one modelled warehouse, so an analyst queries a decade of history as a single table instead of reconciling formats by hand.
- Carried **92** data tests over **17** dbt models, so an upstream schema change breaks the build rather than silently corrupting a published figure.
- Measured disruption impact at **1.42×** median station-day demand, reported as association, not causation, and exposed the warehouse through a read-only MCP server so an LLM agent can query it without credentials.